# Miles Jennings

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## Education

**University of California Irvine** — Irvine, CA
*Bachelor of Science in Computer Engineering* — *Sept. 2022 – June 2026*

**Relevant Coursework**: Organization of Digital Computers, Electronics I–II, Network Analysis I–II, Data Structures and Algorithms, Discrete and Continuous-Time Signals, Digital Logic, Computer Architecture, VLSI, Embedded Systems, Digital Signal Processing, Analog IC Design

**Organizations**: HyperXite, FUSION, FASAE

## Experience

**P2S Technology Intern** — Long Beach, California
*P2S Engineering* — *June 2025 – Present*

- Assisted in the design and documentation of low-voltage and IT infrastructure systems for large scale buildings
- Supported engineers with **CAD** layout updates, wiring diagrams, and equipment schedules using AutoCAD and Revit
- Collaborated with multidisciplinary teams to coordinate technology system requirements with electrical and mechanical designs

**HyperXite – SpaceX Hyperloop** — Irvine, California
*Control Systems Engineer* — *Aug. 2025 – Present*

- Designed, tested, and optimized the Hyperloop pod's control system hardware using an STM32H753ZI microcontroller programmed in **C/C++**, interfacing with thermistors, time-of-flight sensors, and INA219 current sensors while evaluating components for performance and cost improvements over prior pod iterations
- Ensured hardware–software compatibility by defining system constraints and key telemetry parameters based on feedback from team surveys identifying the most critical data points to track in the GUI
- Implemented a Continuous Integration (CI) pipeline using GitHub Actions to streamline testing, version control, and code deployment across subteam

## Projects

**Remote-Controlled Precision Cargo Drone** | *FUSION, Ardupilot* — Dec. 2024 – June 2025

- Designed a UAV using an STM32 based flight controller, incorporating lidar and optical flow sensors capable of **real time data transmission** through UART and ESP32
- Troubleshot PWM output and ELRS radio communication, increasing transmission reliability by 50%

**Hand Gesture Rover** | *Arduino, NRF, ICM* — April 2024

- Designed and built a gesture-controlled rover using Arduino and NRF24L01 modules for wireless communication
- Programmed real-time motion tracking with the ICM's 9-axis IMU to convert hand tilts into directional movement, applying data filtering and SPI-based communication
- Designed and assembled both transmitter and rover hardware, including power circuitry, motor driver integration, and sensor mounting for reliable embedded control

**Building Management System** | *Python, Raspberry Pi, LCD, DHT-11* — April 2025 - June 2025

- Designed and constructed a fully functional Building Management System (BMS) on Raspberry Pi using real-time data from DHT-11 and PIR sensors to control HVAC, ambient lighting, fire alarms, and security systems
- Programmed an interactive LCD interface and logging system to display live status updates, trigger event-based alerts, and ensure time-stamped logs with synchronization and interrupt-driven input handling

## Technical Skills

**Languages**: Python, C/C++, Java, Verilog, RISC-V, MIPS Assembly, MATLAB

**Software/Tools**: Revit, AutoCAD, Bluebeam, Arduino IDE, Vivado, LTSpice, Cadence, Linux, Git, GitHub, STM32CubeIDE, MobaXterm

**Hardware**: Oscilloscope, Multimeter, Circuit Design & Analysis, Microcontrollers, ESP32, Embedded Systems, PCB Design, Arduino, Power Supplies